Approach to

Movement Disorders

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Outlines

- Hyperkinetic vs Hypokinetic movement disorders.
- **Rhythmic** vs **non-rhythmic** movement disorder.
    - Tremor
    - Chorea
    - Athetosis
    - Ballismus
    - Dystonia
    - Myoclonus
    - Tics
- Video presentations of clinical patterns.

Objectives

The students must be able to:

- Recognize and diagnose various patents of presentation of movement disorders.
- Learn the academic definition, the causes and how to investigate for various patents of presentation of movement disorders.

Movement disorders

- Disorders of movement lead to either:
  - Extra, unwanted movement (hyperkinetic disorders: e.g: Hantington's Chorea) or
  - Too little movement (hypokinetic disorders: e.g: Parkinson's D)
- Much of the skill in diagnosing movement disorders lies in pattern recognition.
- Abnormal movements usually imply a disorder in the basal ganglia, in which there is disinhibition of intrinsic rhythm generators or a disorder of postural control.

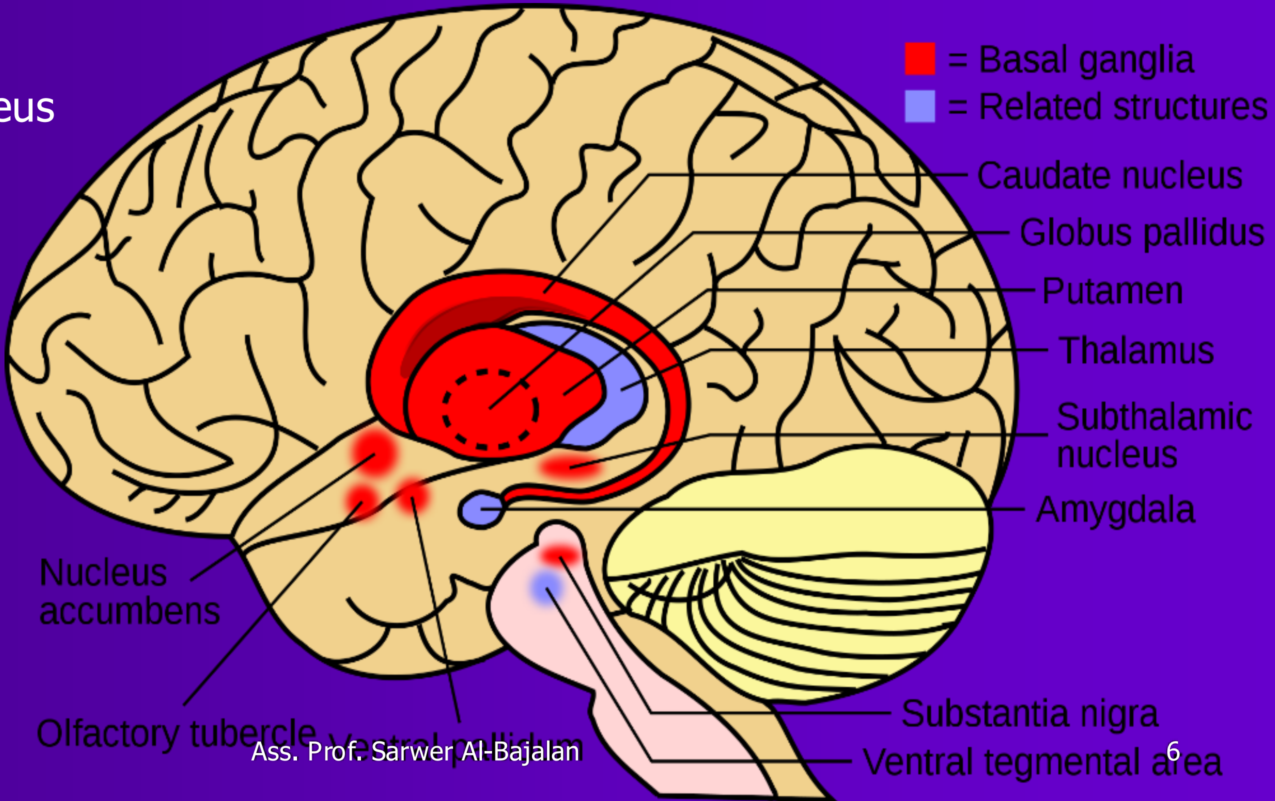
Movement disorders

- Movement disorders present with a wide range of symptoms.
- They may be genetic or acquired, and the most important is Parkinson’s disease.
- Most movement disorders are categorised clinically, with few confirmatory investigations available other than for those with a known gene abnormality.

Anatomy of the Basal Ganglia

The main components are:

- Caudate nucleus
- Putamen
- Olfactory tubercle
- Globus pallidus
- Subthalamic nucleus
- Substantia nigra



25.15 Movement disorders

Description	Features	Examples
Hypokinetic disorders		
Parkinsonism	Akinesia Rigidity Tremor Loss of postural reflexes Other features depending on cause	Idiopathic Parkinson's disease Other degenerative syndromes Drug-induced (See Box 25.54)
Catatonia	Mutism Sustained posturing and waxy flexibility	Usually psychiatric; if neurological, is most commonly of vascular origin
Hyperkinetic disorders		
Tremor	Rhythmical oscillation of body part (see Box 25.16)	Essential tremor Parkinson's disease Drug-induced
Chorea	Jerky, brief, involuntary movements	Huntington's disease Drug-induced
Tics	Stereotyped, repetitive movements, briefly suppressible	Tourette's syndrome
Myoclonus	Shock-like muscle jerks	Epilepsy Hypnic jerks (p. 1086) Focal cortical disease
Dystonia	Sustained muscle contraction causing abnormal postures ± tremor	Genetic Generalised dystonic syndromes Focal dystonias in adults (e.g. torticollis)
Others	Various	Paroxysmal hyperkinetic dyskinesias Hemifacial spasm Tardive syndromes

Tremor

rhythmic movement disorder

Tremor is caused by alternating agonist/antagonist muscle contractions and produces a rhythmical oscillation of the body part affected.

The many causes of tremor have different characteristics:

- **Parkinsonian:** 'Pill-rolling', asymmetrical and present at rest.
- **Physiological:** Exaggerated by anxiety, emotion, drugs , & thyrotoxicosis.
- **Essential:** Slower than physiological tremor; often familial and responsive to propranolol.
- **Dystonic:** Affects head, arms and legs; jerky, associated with dystonias.
- **Functional:** Variable pattern; disappears on distraction.

25.16 Causes and characteristics of tremors

	Body part affected	Position	Frequency	Amplitude	Character
Physiological	Both arms > legs	Posture, movement	High	Small (fine)	Enhanced by anxiety, emotion, drugs, toxins
Parkinsonism	Unilateral or asymmetrical Arm > leg, chin, never head	Rest Postural and re-emergent may occur	Low (3–4 Hz)	Moderate	Typically pill-rolling, thumb and index finger, other features of parkinsonism
Essential tremor	Bilateral arms, head	Movement	High (8–10 Hz)	Low to moderate	Family history; 50% respond to alcohol
Dystonic	Head, arms, legs	Posture	Variable	Variable	Other features of dystonia, often jerky tremors
Functional	Any	Any	Variable	Variable	Distractible

Other hyperkinetic syndromes

non-rhythmic movement disorder

Chorea:

- Jerky, brief, purposeless involuntary movements, appearing as fidgety; they suggest disease in the **caudate** nucleus.

Causes of Chorea include:

- Hereditary: (Huntington's disease, Wilson's disease).
- Drugs: (levodopa, antipsychotics, anticonvulsants*, oral contraceptives).
- Autoimmune: Sydenham's chorea, antiphospholipid syndrome, SLE.
- Endocrine: pregnancy, thyrotoxicosis, hypoglycemia & hypoparathyroidism.
- Structural lesions: vascular, demyelination, brain tumors.

Huntington's Chorea

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Athetosis:

Slower writhing movements of the limbs. These are often combined with chorea and have similar causes.

Ballism:

- Flinging movements of the limbs more dramatic than chorea.
- Usually occur unilaterally (**hemiballismus**) in vascular lesions of the **subthalamic nucleus.**

Dystonia:

- A movement disorder in which a limb (or the head) involuntarily takes up an abnormal posture.
- This may be generalized in basal ganglia disease, or may be focal or segmental, as in **spasmodic torticollis** when the head involuntarily turns to one side.
- Other segmental dystonias may cause abnormal disabling postures during specific actions (e.g. **writer's cramp**) and can be treated with **botulinum toxin** to the responsible muscles.

Athtosis in a CP patient

Hemiballismus

Generalized Dystonia

Acute Drug Induced Dystonia

Newcastle University

Severe Cervical Dystonia

Writer's Cramp

Myoclonus:

Myoclonus is a brief, isolated, random (non-rhythmic), non-purposeful jerks of muscle groups in the limbs.

- They occur normally at the onset of sleep (hypnic jerks).
- Similarly, a myoclonic jerk is a component of the normal startle response, which may be exaggerated in some rare (mostly genetic) disorders.
- Myoclonus may occur in disorders of the cerebral cortex, such as some forms of epilepsy.
- Alternatively, myoclonus can arise from subcortical structures or, more rarely, from segments of the spinal cord.

Epileptic myoclonus (JME)

myoclonic seizure

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Propriospinal Myoclonus

Which of the following statements about tic disorders is true?

A) Isolated tics are more common in adulthood than in childhood.
B) Patients with tic disorders are unable to suppress their tics.
B) Tics can only be simple, not complex.
D) Tics can be a symptom of Huntington's and Wilson's diseases.

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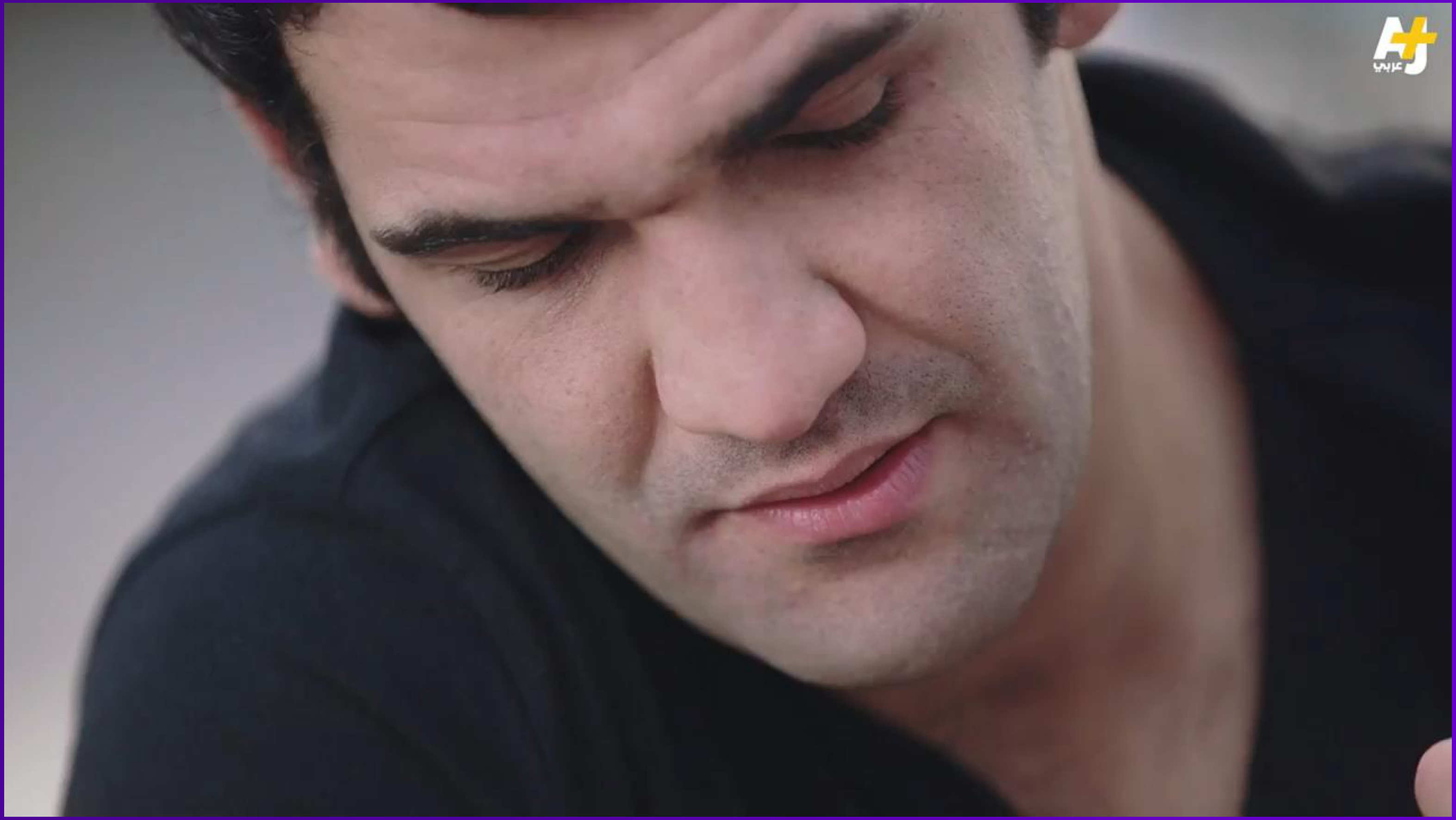
Answer D) Tics can be a symptom of Huntington's and Wilson's diseases.

Tics:

- Repetitive semi-purposeful movements, such as blinking, winking, grinning or screwing up of the eyes.
- Unlike with other involuntary movements, patients can suppress tics, at least for a short time.
- Isolated tics are common in childhood and usually disappear.
- Tics can be simple or complex.
- **Tourette's syndrome** is defined by the presence of multiple motor and vocal tics that may evolve over time; it is frequently associated with psychiatric disease, including:
  - obsessive compulsions,
  - depression,
  - self-harm or
  - attention deficit disorder.
- Tics may also occur in Huntington's and Wilson's diseases, or after streptococcal infection.

A Child with Tic Disorder

Tourette's syndrome



Concusions

- Video presentations make understanding of the patterns of movement disorders very much easier.
- Rest tremor is characteristic of PD.
- Hemiballismus is the must aggressive type of movement disorder that caused by vascular insults.
- Chorea & athetosis are frequently combined together & have simillar wide range of causes.
- Torrent syndrome is an uncommon tic disorder associated with vocalization & may be disabling in some case.


Thanks A lot for your attention